Rafael Arturo Coca Salazar

rafael.coca.sal@gmail.com

Education

Princeton University **Princeton, NJ**
Bachelor of Arts in Physics; Minor in Mathematics *Sep. 2021 – May 2025*
GPA: 3.47/4.0

Research Experience

Lawrence Berkeley National Laboratory **Remote**
Beam-induced background studies with USMCC *Sep. 2025 – Present*
- Investigated alternate oversampling technique keeping particle correlations with no appreciable difference.
- Currently working on the design and implementation of a machine learning model to generate any proportion of background data training from the current fraction of simulated data.

Princeton University **Princeton, NJ**
2+1D QED Hamiltonian Lattice Model *Sep. 2024 – Apr. 2025*
- Obtained the 2+1D QED Hamiltonian lattice prescription based on previous work and characterized its representations.
- Implemented the model using Mathematica's efficient computation methods.
- Computed ground state energy, fermion bilinear, and mass gaps on a 2×2 lattice, reproducing perturbative results.
- Inferred Hilbert-space scaling for larger lattices based on an electric-field cutoff.

CERN **Geneva, Switzerland**
Analysis of Level-1 Trigger Algorithm *Jun. 2024 – Jul. 2024*
- Investigated the CICADA algorithm of the CMS Level-1 Trigger by generating deposit patterns.
- Identified minor systematic biases attributed to miscalibrated calorimeter towers.
- Worked on generalizing the analysis for ML-based trigger algorithms in other experiments.
- Gained experience with CMS detectors and the CERN data analysis pipeline.

Princeton University **Princeton, NJ**
Large Order Behavior of QM Correlation Functions *Jan. 2024 – Apr. 2024*
- Derived the instanton calculation for 0+1D spacetime.
- Estimated growth and convergence of correlation functions using a toy model.
- Obtained original results for one-point functions and large-order behavior to arbitrary order.
- Simplified and condensed calculations for two-point functions to second order.

Princeton University **Princeton, NJ**
Review on Axion Detection Experiments *Sep. 2023 – Jan. 2024*
- Reviewed modern axion detection experiments for an undergraduate audience.
- Emphasized the role of sensitivity and amplification in precision experiments.
- Developed familiarity with axion phenomenology and detection theory.

Princeton University **Princeton, NJ**
Heading Error Corrections in SERF Magnetometers *May 2023 – Jul. 2023*
- Probed heading errors present in SERF magnetometers.
- Identified and isolated errors proportional to the generated magnetic field.
- Designed and built proof-of-concept Helmholtz coils to correct or reduce heading errors.

Helmholtz Zentrum Berlin **Berlin, Germany**
Design and Implementation of Follow Focus System *May 2022 – Aug. 2022*
- Iterated through design of camera chassis for proper mounting and operation.
- Programmed an interface for remote operation of the follow-focus system.

Relevant Research Skills

Theoretical & Computational Modeling: Derived and implemented lattice QED Hamiltonians in Mathematica; computed ground state energy, fermion bilinears, and mass gaps.

Mathematical Methods: Applied instanton calculus, path integrals, and large-order perturbation theory to quantum mechanical systems.

Data Analysis & Simulation: Analyzed calorimeter tower behavior and systematic biases using ROOT; experience with TensorFlow/Keras for HEP ML workflows.

Teaching Experience

Princeton University **Princeton, NJ**
Undergraduate Course Assistant *Feb. 2023 – May 2024*
- Hosted office hours, led problem sessions, and prepared solution sets.
- Courses: PHY103 Introductory Mechanics; PHY104 Introductory Electromagnetism.

Princeton University **Princeton, NJ**
Tutor, Physics Department Peer Tutoring *Sep. 2023 – Dec. 2024*
- Tutored students in problem-solving, conceptual reasoning, and analytical skills.
- Subjects: Classical Mechanics; Quantum Mechanics I & II.

Additional Skills

Coding: C++, ROOT, Python, TensorFlow, Keras, cluster usage, Mathematica, Java

Tools: Git/GitHub, Linux/Unix, LaTeX, Creo Parametric, NX Siemens

Languages: English (C2), Spanish (Native)

Extracurricular Activities

Princeton University **Princeton, NJ**
President, Princeton Society of Physics Students *Feb. 2024 – Jan. 2025*
- Increased engagement in undergraduate physics community events.
- Hosted accessible research talks for undergraduate and graduate audiences.
- Managed mentoring program and curated academic/career resources.
- Contributed problems to the Physics Unlimited Premier Competition 2024.

Princeton University **Princeton, NJ**
Treasurer, Towards an Inclusive Community of Undergraduate Physicists *Feb. 2023 – Jan. 2024*
- Helped secure funds for events supporting undergraduate physicists.
- Raised mental-health awareness via events and newsletters.

Membership & Awards

2022–2025: American Physical Society

2018, 2019–2020: **Bolivian Ministry of Education, Olympiad Finalist**: Bronze medals (intermediate and advanced levels) in the Bolivian National Computer Science Olympiad; trained and led high-school team.

Referees

Prof. Silviu Pufu — `spufu@princeton.edu`

Prof. Isobel Ojalvo — `iojalvo@princeton.edu`

Dr. Simone Pagan Griso — `spagangriso@lbl.gov`